

Jordan Buckley

Jordanbuckleycork@gmail.com | +353 894476856 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

University College Cork - Computer Science (CK401)

• 1st Year Class Representative

September 2024- September 2028

Grade - 1.1 First Class Honors (Among the top performers in the year)

Core Modules (2nd Year) :

- Algorithms & Data Structures II | Intermediate Programming | Cloud Infrastructure & Services
- Computer Architecture | Information Storage & Management II | Operating Systems

Professional Experience

Founding Engineer

Barrikade | Python, TypeScript, Tailwind CSS, Postman, Git

April 2026 – Current

- Architecting the registry and credential system for agent-to-agent authentication, designing a SPIFFE-based identity model to verify and authorize autonomous AI agents at runtime.
- Built and shipped [\[Barrikade.ai\]](#) (Next.js, Typescript, Tailwind CSS, Loops API), including manifest-based site indexing so LLM-driven search agents could discover the product.
- Wrote a Postman test suite to validate API endpoint behaviour ahead of deployment, catching integration issues pre-release.
- Exploring Model Context Protocol (MCP) integration to extend secure tool-calling to LLM agents.

Google BGN Hackathon Winner

Google, Dublin

October 2025 – October 2025

SolarMatch | Next.js, Typescript, Tailwind CSS, FastAPI, Python, Docker, PostgreSQL, Firebase, GCP

- Won 1st place for developing *SolarMatch*, Ireland's first solar assessment and cooperative platform.
- Trained a Gemini chatbot on SEAI data for automated grant eligibility, reducing user research time by 70%.
- Engineered financial models in Python/FastAPI with $\pm 5\%$ ROI accuracy and automated cost estimation.
- Designed a B2B/B2G revenue model leveraging affiliate links with solar providers and government incentives for referrals.
- Built scalable APIs integrating Gemini, SolarAPI, and Google Maps, achieving 30% higher yield accuracy via satellite heatmap modelling. [\[GitHub\]](#)

Webmaster

UCC ACM Student Chapter | JavaScript, HTML, CSS, SSH, Docker

September 2024 – Current

- Rebuilt and optimized the chapter website, improving load speeds by 55% and uptime to 99.9%.
- Designed and deployed scalable server infrastructure; documented architecture for future scalability.
- Managed sponsor visibility for hackathons with F5, Cadence, Keelvar, Insight, and NetApp, increasing engagement by 40%.
- Automated deployments via CI/CD, cutting update time by 70%. [\[UCC ACM\]](#)

Projects

Grid Queue Simulator | Java

- Architected an extensible capacity allocation simulator using Strategy and Factory design patterns to model substation load allocation across competing industrial and residential demands.
- Engineered 4 interchangeable priority algorithms (FCFS, Housing-First, Priority-Based, Capacity-Weighted) across a modular 6-layer pipeline to benchmark workload CSV traces. [\[GitHub\]](#)

Black-Scholes Option Pricing Model | Python, Streamlit, NumPy, SciPy

- Engineered quantitative pricing modules calculating European call/put valuations and analytical Greeks (Delta, Gamma, Vega, Theta, Rho) via SciPy cumulative normal distribution algorithms.
- Built matrix-based heatmaps and automated reporting using NumPy, Seaborn, and dynamic FPDF generation to render real-time P&L payoff curves and export analytical PDF summaries. [\[BlackScholes\]](#)

Movie DNA | Python, Flask, SQLite, SQL, Werkzeug

- Engineered a content-based recommendation algorithm aggregating watchlist history and genre profiles to calculate dynamic user movie matches.
- Built parameterized SQLite schemas with complex JOIN aggregations, securing endpoints via PBKDF2 password hashing, session cookies, and Role-Based Access Control (RBAC). [\[GitHub\]](#)

Skills

Languages: Python, C++, Java, C, Javascript, Typescript, HTML/CSS, SQL, Bash

Frameworks & Libraries: React.js, Flask, Node.js, Tailwind CSS, FastAPI, Next.js, Pandas, NumPy, Matplotlib, Firebase, Streamlit

Databases: PostgreSQL, MongoDB, Neo4j, SQLite

Tools & Cloud: AWS, GCP, Kubernetes, Docker, Git, Postman, CI/CD, SSH